PM518a Project

Date Assigned: Apr 8, 2026
Date Due: May 10, 2026

Background: The Framingham cohort study is a long-term prospective study, examining determinants of cardiovascular disease in a free-living population of residents of the city of Framingham, Massachusetts. The study began in 1948. Cardiovascular disease outcomes were identified through routine surveillance of area hospitals, participant reports, and death certificates. For more information, go to https://www.nhlbi.nih.gov/science/framingham-heart-study-fhs

Dataset: All subjects began follow-up for incident outcomes at the baseline exam. Subjects with prevalent coronary heart disease (CHD, at the baseline exam) were *excluded* from this dataset. The primary outcome of interest in this analysis is time to first nonfatal myocardial infarction (MI) or coronary death *since* baseline examination.

Objective: Given the growing prevalence of diabetes in industrialized nations, the investigators are particularly interested in the contribution of diabetes (`diabetes`) to MI/CHD risk (`mi_fchd`). As a secondary hypothesis, the investigators are also interested in the contribution of cholesterol levels (`totchol`) to MI/CHD risk. Both diabetes status and cholesterol levels are modifiable risk factors (i.e., factors that one might change through lifestyle or use of medications) for coronary heart disease. Other modifiable risk factors are included in the data, but our focus will only be on diabetes and cholesterol levels. Non-modifiable variables in the dataset that might be related to coronary heart disease risk as well as the modifiable risk factors above include age (`age`) and sex (`sex`), with education (`educ`) as an additional variable for possible adjustment.

General instructions:

1. Besides your group partner, you may not discuss strategies with students or anyone else, including tutors or the TA. Dr. Kawaguchi will answer any questions of clarification you might have. Evidence that you have used outside help in any way will result in zero credit for your final project, a grade of F for the class, and a report describing the incident submitted to your department chair and to the graduate school. Evidence of shared work between two or more groups will result in the same penalties for all involved students. Your project will not be graded unless you have signed and turned in the signed <u>Certification of Academic Integrity</u> form (also due at the same time as the Final Project).

2. The project should be turned in through Brightspace by the due date (May 10th, 11:59pm PDT). **<u>Do not</u>** wait until the last second to submit your assignment. Any assignment received *after* the deadline based off *Brightspace's internal clock* will be subject to a score of **<u>0</u>**.

3. Turn in two project files:
    a. PDF file of your final project report that includes text, tables, and figures: Name this file *"pm518a_project_text_ group_[group number].pdf"*
    This document should be no longer than 10 pages (including tables and figures), using Arial Font 11 single spaced. Any documents longer than 10 pages will not be accepted. Ten pages is a generous, I do not expect your projects to reach this page limit.
    Note: Both groups members should be turning in the **same** report. The TA will only be grading <u>one</u> of the two students' uploaded documents. If one student uploads an incomplete writeup and the TA happens to grades that one, the Final Score will be based off of that writeup. Make sure both writeups are consistent. While this is a group project, both members are **expected** to work on ALL questions.

    b. Your Stata do file: Name this file *"pm518aproject_smcl_group_[group number].do".*
    I should be able to run your Stata do file to obtain the results shown on your report (with very minor modifications). **Note:** Both you and your group partner should turn in the *same* .do file.

Your final project report should read like a journal article (see examples that I provided in class as well as in journals such as JAMA, American Journal of Epidemiology, International Journal of Epidemiology, etc.).

Specifically, your report should include a **Data**, **Methods**, **Results**, and **Discussion** section. Also include a **Supplemental materials** section at the end. **Tables and figures must be of publication quality.** You will be scored off for tables and figures that are not of publication quality. For example, figures and tables should be properly labeled. Dataset variable names (e.g., `totchol`) should not be used. Do not include more than two decimal places. Stata output *should not* be directly used as a table for your results. Be sure to include captions to your tables and figures and reference them within the text when discussing your results (tables and figures in the supplemental should start with S, e.g., Table S1, Figure S1, etc.). Lastly, note that you do **not** include results or interpretations of your findings/analysis in the Method section. If you are unsure about how to write any of these methods, I advise you to read relevant journal articles.

Assignment:

1. You must request access to the Framingham Heart Study data through BioLINCC (https://biolincc.nhlbi.nih.gov/teaching/).

    a. If you do not already have an account, please create one using your @usc.edu email.

    Request Name: "PM518a Final Project"
    Requester Name: Your name
    Institution: University of Southern California
    Email: Use your USC email address
    Intended Use: "Student Use"
    Dataset: Framingham

    b. Please forward **Alexander** the email you get from Biolincc that shows you have proper access to the data. The body of the email might look something like:

    "

    The status of BioLINCC request "PM518a Use of Framingham Dataset" was changed from "BioLINCC Review" to "Fulfilled" by XXXXX.

    Please review the comments tab for instructions and/or a link to access the dataset.

    You may access the request here: https://biolincc.nhlbi.nih.gov/requests/teaching-dataset-request/...../
    "

    Note: While you may be able to get the Stata file (.dta) from your classmates, this is a **mandatory** step for the final project. Failing to email the TA with this information *before* submitting your final project will count as improper data usage and lead to a score of **0** for your Final Project.

2. Before starting your analyses:

    a. The event of interest is "Hospitalized myocardial infarction or fatal coronary heart disease" which can be found under `mi_fchd`. The corresponding time-to-event is `timemifc`, which is measured in days since baseline. Please convert the time scale from days to years by creating a new variable that is equal to `timemifc/365.25`.
    b. Confine your analysis to baseline measurements of risk factors (`period = 1`).
    c. Remove individuals who have a survival time of zero from the analysis.
    d. Draft a **Data** section that briefly describes the study data. Include the total number of observations you start with and how many are excluded from Part c (and why). Discuss any modifications you made to the original dataset.

    *Your sample size should be 4,354 individuals

3. Using a significance level of $\alpha = 0.05$, draft a **Methods** (in particular, a Statistical Analysis subsection) section, and a **Results** section that include the following:

   a. Examine the distributions of your independent variables by developing a descriptive table (i.e., Table 1), with separate columns for subjects with and without diabetes. Include the number of missing values, if any. Remember to include units of measurement in your table.

   b. Provide a Kaplan-Meier survival curve, comparing persons with and without diabetes.
      i. Format your figure with an overall title, titles on the x- and y-axes, and appropriate legend labels.
      ii. Include a risk table in your figure that indicates the numbers at risk over follow-up at: 0, 5, 10, 15, and 20 years.
      iii. Place your legend inside the figure

   c. In each group (diabetics and non-diabetics):
      i. Estimate median survival along with 95% confidence intervals.
      ii. Estimate 5-, 10-, 15-, and 20-year survival, along with 95% confidence intervals.
      iii. In addition to (ii), test if 5-, 10-, 15-, and 20-year survival differs between the two groups. Since you are performing five tests, use a Bonferroni-corrected alpha level, $\alpha^* = 0.05/5 = 0.01$.

   d. Test for differences in the two survival curves using a log-rank test.

   e. Using a Cox proportional hazards model:
      i. Estimate and test the univariate association of diabetes and of total cholesterol levels on risk of MI/CHD.
         1. Due to the wide range of total cholesterol levels, center your cholesterol levels around the sample mean and <u>express per 10 unit increase</u> when modeling. Be sure to describe this change somewhere in your article.
      ii. For total cholesterol levels:
         1. Use martingale residuals to evaluate the linearity assumption.
         2. Provide lowess plots of martingale residuals.
         3. Discuss if any transformation is needed. If so, perform transformations to achieve linearity.
         4. Include any relevant figures or tables in the **Supplemental materials** section.

   f. Develop a multivariable main effects model with the two modifiable risk factors of interest. Consider whether age, sex, and education are possible confounders.
      i. Test the proportional hazards assumption for your multivariable model. Identify any variables that do not meet proportional hazards. If needed, modify your modeling to deal with this issue. Include any relevant figures or tables in the **Supplemental materials** section.
      ii. Complete model diagnostics on your final multivariable model. Include any relevant figures or tables in the **Supplemental materials** section.

iii. Make a table of your univariate and multivariable associations, including HRs, 95% confidence intervals and p-values.

g. A secondary hypothesis that the investigators have is to test whether the total cholesterol association with MI/CHD hazard varies by diabetes status. Provide a test on the hypothesis.

h. A fellow co-author suggested you perform a secondary analysis by categorizing cholesterol levels as follows: < 200 mg/dl (healthy), 200 – 239 mg/dl (at risk), 240+ (dangerous). Estimate and test both the unadjusted and adjusted associations of cholesterol (categorical) on risk of MI/CHD. You can use the same multivariable model as in Part f. Include any figures and tables based off of this analysis in the **Supplemental materials** section.

4. In, at most, a few paragraphs, summarize your findings and discuss any potential limitations to the current study (e.g., a small **Discussion** section).

5. At the end, create a small **Contributions** section that describes/lists the contributions of both **you and your group partner** for this final project.